

Can RL Teach Long-Horizon Reasoning to LLMs? Expressiveness Is Key

Tianle Wang¹ Zhaoyang Wang² Guangchen Lan¹ Xinpeng Wei³
Sipeng Zhang⁴ Guanwen Qiu¹ Abulhair Saparov¹

¹Purdue University ²UNC Chapel Hill ³Georgia Tech ⁴UC San Diego
wang7327@purdue.edu

Abstract

Reinforcement learning (RL) has been applied to improve large language model (LLM) reasoning, yet the systematic study of how training scales with task difficulty has been hampered by the lack of controlled, scalable environments. We introduce SCALELOGIC, a synthetic logical reasoning framework that offers independent control over two axes of difficulty: the depth of the required proof planning (i.e., the *horizon*) and the expressiveness of the underlying logic. Our proposed framework supports a wide range of logics: from simple implication-only logic (“if-then”) towards more expressive first-order reasoning with conjunction (“and”), disjunction (“or”), negation (“not”), and universal quantification (“for all”). Using this framework, we show that the RL training compute T follows a power law with respect to reasoning depth D — $T \propto D^\gamma$, $R^2 > 0.99$ —and that the scaling exponent γ increases monotonically with logical expressiveness, from 1.04 to 2.60. On downstream mathematics and general reasoning benchmarks, more expressive training settings yield both larger performance gains (up to +10.66 points) and more compute-efficient transfer compared to less expressive settings, demonstrating that *what* a model is trained on, not just *how much* it is trained, shapes downstream transfer. We further show that the power-law relationship holds across multiple RL methods, and curriculum-based training substantially improves scaling efficiency.

1 Introduction

Recent advances in reinforcement learning (RL) for large language model (LLM) reasoning have shown that training on verifiable domains such as mathematics and coding can elicit enhanced reasoning behavior and improve benchmark performance [Guo et al., 2025a, Jaech et al., 2024]. However, these gains do not yet translate into robust long-horizon reasoning, as the same model’s performance can degrade sharply once tasks require many sequential reasoning steps, even when the corresponding subproblems are individually manageable [Rameshkumar et al., 2025, Motwani et al., 2025, Lu et al., 2025, Zhou et al., 2025].

A key limitation is that scalable long-horizon reasoning training requires three properties simultaneously: exact verifiability, fine-grained control over reasoning difficulty, and data available at scale that supports systematic analysis. The domains currently driving RL progress, such as mathematics and coding, often satisfy the first property but not the latter two: high-quality problems are expensive to curate and offer only limited control over horizon and difficulty [Hendrycks et al., 2021, Jain et al., 2024]. Existing alternatives, including synthetic tasks [Xie et al., 2025, Liu et al., 2025a] and self-evolving pipelines [Zhao et al., 2025, Huang et al., 2025], reduce the cost of data collection through automatic generation and verification, but often do not provide a sufficiently clean and interpretable set of control axes for analyzing scaling behavior. As a result, we still lack a controlled training setting for studying how RL scales with increasingly difficult long-horizon reasoning problems. Table 1 summarizes how existing data sources fall short on at least one of these axes.

Table 1: Comparison of training data sources for RL post-training of LLM reasoning. Here, ✓ denotes direct support, ◦ denotes coarse or indirect control, and × denotes no direct control; e.g., SAT can vary the number of variables and clauses, but only indirectly controls the reasoning horizon.

Data Source	Verifiable	Scalable	Controllable Horizon	Controllable Expressiveness
Math / Code [e.g., Hendrycks et al., 2021]	✓	×	×	×
Game-based [e.g., Wu et al., 2024]	◦	✓	×	×
Self-Evolving [e.g., Zhao et al., 2025]	✓	✓	×	×
Knights and Knaves [Xie et al., 2025]	✓	✓	◦	×
SAT [Liu et al., 2025a]	✓	✓	◦	×
G1 [Guo et al., 2025b]	✓	✓	◦	×
Puzzle Suites [e.g., Liu et al., 2025b]	✓	✓	◦	×
SCALELOGIC	✓	✓	✓	✓

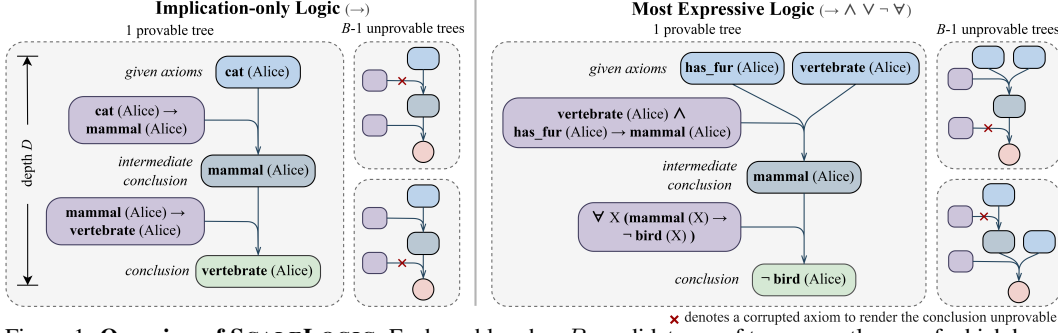
To address this gap, we propose SCALELOGIC, a synthetic logical reasoning environment with explicit control over two key dimensions of difficulty: the depth of the required proof planning (i.e., the *horizon*) and the logical expressiveness of the reasoning problems. As illustrated in Figure 1, in each example, the model is given a set of facts and a handful of candidate conclusions. The model is then asked to identify which candidate conclusion is logically derivable from those facts, yielding an exact and easily verifiable multiple-choice objective. To find the correct multiple-choice option, the model must search for a proof of each candidate conclusion. Our environment requires it to expend comparable reasoning effort to determine the provability of each candidate, which substantially limits the model’s ability to exploit simple heuristics or shortcuts during reasoning. Beyond scaling the depth of the proof tree, our environment allows us to vary the expressive structure of the underlying logic, including the logical operators and structural properties involved in each example. Combined with automatic low-cost generation and verification, this provides a controlled testbed for systematically studying RL scaling in long-horizon reasoning.

Experiments in this controlled environment show that over the observed depth range, the RL training steps T required to reach 90% validation accuracy follow a power law in the proof-tree depth D ($T \propto D^\gamma$, $R^2 > 0.99$), where the scaling exponent γ increases monotonically with logical expressiveness, from 1.04 to 2.60. This indicates that more logically expressive settings require disproportionately more training to solve problems of the same depth. We also find that the power-law relationship holds across different RL methods, and these dynamics depend strongly on the training distribution: a carefully designed curriculum improves training efficiency and stability.

We also demonstrate that training a model in our synthetic reasoning environment improves performance on downstream real-world mathematics and general reasoning benchmarks, with the most expressive setting improving the mean accuracy across eight benchmarks by up to +10.66 percentage points over the base model. Critically, training on highly expressive data yields a monotonically increasing downstream performance curve, while less expressive settings plateau early. This indicates that *what* the model is trained on, not just *how much* it is trained, shapes downstream transfer. Taken together, these results suggest that the logical expressiveness of the training data influences not only how RL scales on synthetic reasoning tasks, but also the extent to which such training can provide transferable improvement in reasoning performance for real-world applications.

In summary, we introduce a scalable synthetic reasoning framework for studying and improving long-horizon reasoning in RL post-training. Our main contributions are as follows:

- We propose SCALELOGIC, a controlled and scalable synthetic logical reasoning testbed for RL post-training, with exact verifiability, low-cost automatic generation, and explicit control over reasoning horizon and logical expressiveness.
- We show that RL training effort follows a power law with respect to proof tree depth, with the scaling exponent γ increasing monotonically with logical expressiveness, from 1.04 to 2.60. The power-law relationship holds across multiple RL methods, and we further show that curriculum training improves scaling efficiency.
- We show that synthetic reasoning training improves downstream reasoning performance by up to +10.66 points, and these gains depend strongly on the logical expressiveness of the training environment, with more expressive settings yielding larger and more compute-efficient transfer.



2 Related Work

Long-horizon Reasoning Limitations. Recent work reveals that LLMs often exhibit sharp performance degradation as the reasoning horizon increases. Rameshkumar et al. [2025] show that reasoning models can perform well on graph-based reasoning tasks within a limited complexity regime, but their performance drops abruptly once the reasoning horizon is sufficiently large. SeqBench [Ramezani et al., 2025] and GSM-Infinite [Zhou et al., 2025] also report steep performance degradation with exponential and sigmoid-like decay patterns. Further studies, such as R-Horizon [Lu et al., 2025] and h1 [Motwani et al., 2025], compose individually solvable mathematical problems into multi-step dependency chains and investigate whether RL training can mitigate long-horizon failures. Building on this line of work, we characterize how RL training behaves as reasoning structures are systematically scaled, and how these scaling dynamics shape downstream transfer.

Scaling in LLMs. Early scaling-law studies found regular power laws relating pre-training performance to model scale, data volume, and training compute [Kaplan et al., 2020, Henighan et al., 2020, Hoffmann et al., 2022]. Later, test-time scaling emerged to improve reasoning by allocating additional computation during decoding [Wei et al., 2022, Wang et al., 2023, Yao et al., 2023, Muennighoff et al., 2025]. Recent work has extended scaling-law analysis to RL post-training, showing regular scaling behavior with model scale, data, and compute [Khatri et al., 2025, Tan et al., 2025]. However, existing RL scaling studies primarily vary the volume of training data, while providing limited control over the reasoning complexity of individual problems. In contrast, our work enables a cleaner analysis through explicit and interpretable control of reasoning complexity.

RL for LLM Reasoning. RL with verifiable rewards (RLVR) has become a promising paradigm for reasoning-oriented post-training [Luong et al., 2024, Guo et al., 2025a]. With practical optimizers such as GRPO [Shao et al., 2024] and later variants [Yu et al., 2025, Zheng et al., 2025], it has enabled large-scale RL training and long chain-of-thought reasoning [Jaech et al., 2024, Guo et al., 2025a]. However, most existing RL reasoning work centers on mathematics and programming, where high-quality training problems are limited, often depend on human-curated solutions or test cases, and offer only coarse difficulty control [Liu et al., 2025a]. As a result, RL performance is limited by the collected data, making sustained improvement difficult as training scales. In contrast, SCALELOGIC offers explicit complexity control, verifiable solutions, and unlimited low-cost data generation, enabling a cleaner and more scalable framework for reasoning-oriented RL.

Synthetic Data for Post-training. Synthetic data is a natural fit for RLVR, motivating recent work on automatically generated reasoning problems with verifiable rewards. Early efforts studied task-specific synthetic settings for logical reasoning, such as Knights and Knaves [Xie et al., 2025, Lin et al., 2025]. More recent work shifted toward task families with difficulty control such as SAT [Liu et al., 2025a], graph reasoning tasks such as G1 [Guo et al., 2025b], and benchmark-style synthetic reasoning suites [Liu et al., 2025b, Chen et al., 2025, Stojanovski et al., 2025, He et al., 2026]. While these studies show the promise of synthetic data for post-training, their tasks are typically worst-case NP-hard search problems (e.g., SAT and Hamiltonian path in G1) with limited control over underlying expressiveness, making it difficult to isolate how task complexity affects RL training compute and downstream reasoning. In contrast, SCALELOGIC independently controls proof depth and logical expressiveness with each instance oracle-solvable in time linear in the proof size.